

Technology Stack

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Bootstrap, Jinja2	Bootstrap ensures a responsive, clean, and mobile-friendly UI. Jinja2 is used as the templating engine to dynamically render data and visualizations directly from the Flask backend.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask Framework	Python is the industry standard for Data Science and Machine Learning. Flask is a lightweight, flexible micro-framework that seamlessly integrates with ML libraries and handles API routing efficiently.

3	Database / Data Storage (e.g., <i>MySQL, MongoDB, PostgreSQL</i>)	SQLite (via SQLAlchemy) & CSV (Pandas)	SQLite provides a lightweight, serverless relational database to store user profiles and prediction history. CSV/Pandas DataFrames are used to efficiently store, query, and process the structured UNDP HDI dataset.
4	Cloud / Hosting / Deployment (e.g., <i>AWS, Azure, Heroku, Docker</i>)	Docker & Render / Heroku	Docker containerizes the application to ensure it runs consistently across all environments. Render/Heroku provides easy, cost-effective, and scalable cloud hosting specifically optimized for Python/Flask apps.
5	Version Control & CI/CD (e.g., <i>GitHub, GitLab, Jenkins</i>)	GitHub & GitHub Actions	GitHub enables seamless team collaboration, code reviews, and version tracking. GitHub Actions automates the CI/CD pipeline (testing and deployment) to ensure code quality and smooth releases.
6	Third-Party APIs / Other Tools (e.g., <i>Stripe, Twilio, Google Maps</i>)	Scikit-learn, Pandas, NumPy, Matplotlib/Seaborn, World Bank / UNDP API	Scikit-learn powers the core ML prediction engine. Pandas/NumPy handle data manipulation. Matplotlib/Seaborn generate EDA charts. The UNDP/World Bank API fetches real-time global development statistics.